

EC6020

Embedded Systems Design

Project Proposal



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TITLE: Roadside Fog Density Monitoring & Accident Prevention System

A low-cost embedded system for real-time fog detection and warning system using infrared sensing and long-range wireless communication.

INTRODUCTION

Fog significantly reduces visibility and is a major cause of accidents on highways and mountain roads, especially on winding routes with sharp curves and elevation changes. In many rural and hilly regions, there are no automated systems to detect fog and warn drivers in advance, while existing solutions are often expensive or infrastructure-heavy.

This project proposes FogGuard, an embedded fog detection and early warning system using infrared sensing, environmental monitoring, and LoRa communication. The system detects fog intensity and transmits warnings to stations up to 1 km ahead, providing real-time alerts through different-colored LEDs and an LCD display to help drivers reduce speed and improve safety in low-visibility conditions.

DESIGN OVERVIEW

(01) System Architecture:

The system consists of two main units:

1. Fog Detection Unit (Transmitter Node)

- Installed at fog-prone highways and mountain roads, especially on winding routes locations
- Measures fog intensity using IR scattering
- Sends fog level data wirelessly using LoRa

2. Warning Display Unit (Receiver Node)

- Installed approximately 1 km ahead
- Receives fog data
- Displays warning lights
- LCD Display

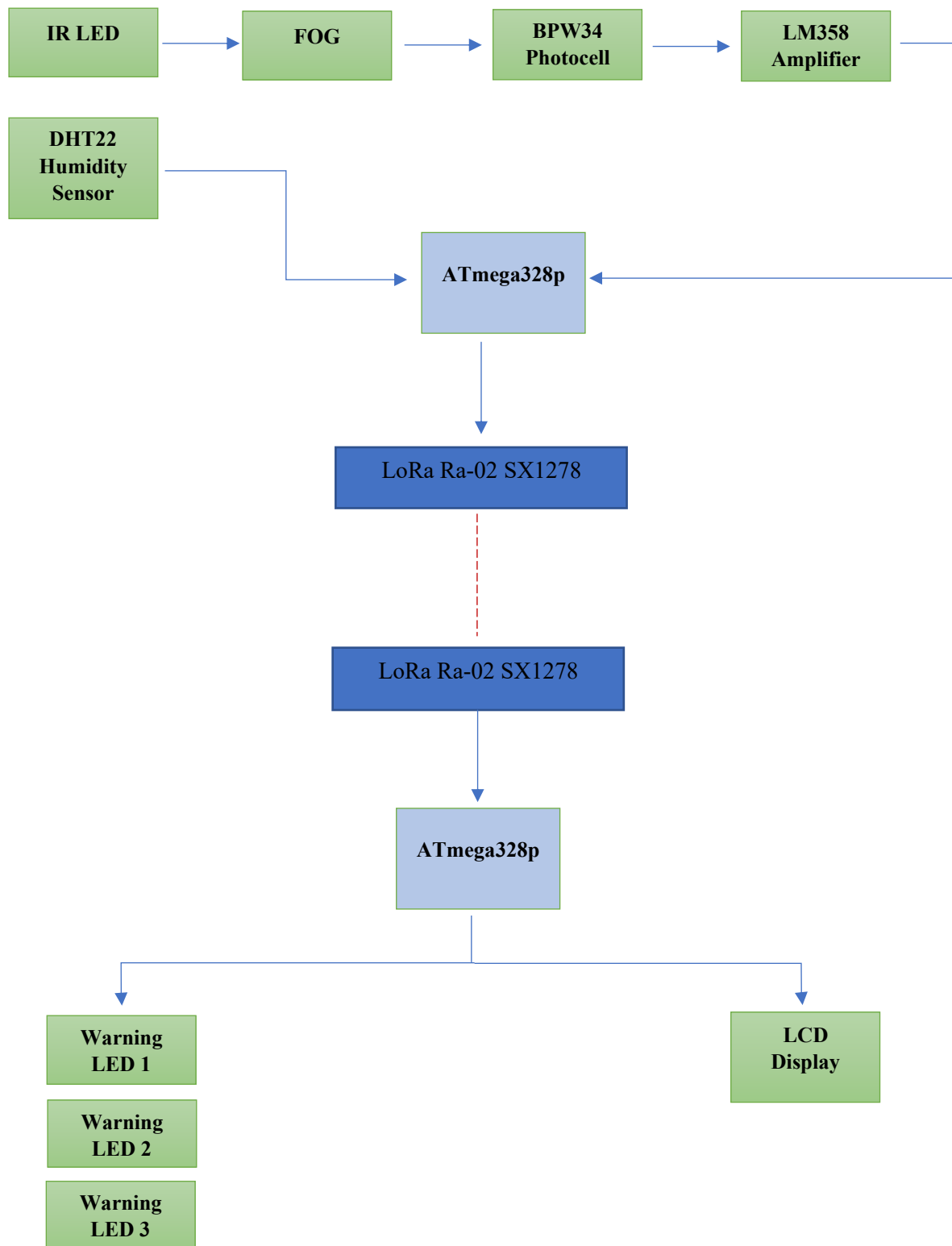


FIGURE 01: SYSTEM ARCHITECTURE

(02) Protocols:

1. Serial Peripheral Interface (SPI)

SPI is used for communication between the ATmega32 microcontroller and the RA-02 (SX1278) LoRa module. SPI provides high-speed, full-duplex data transfer, which is essential for sending sensor data and control commands efficiently. It is natively supported by the ATmega32 hardware, ensuring reliable communication with minimal software overhead.

2. Single-Wire Protocol (DHT22 Communication)

The DHT22 temperature and humidity sensor uses a proprietary single-wire communication protocol based on precise timing. The microcontroller initiates data transmission, and the sensor responds with calibrated humidity and temperature values.

3. GPIO (General Purpose Input/Output)

GPIO pins are used for LED control, IR emitter modulation, and digital signal handling. Different LED colors indicate fog severity levels, while GPIO control ensures fast and deterministic operation.

4. LoRa PHY (Physical Layer Communication)

The LoRa physical layer (PHY) is used for long-range wireless communication between the fog detection unit and the warning station. LoRa enables communication over distances up to 1 km or more with very low power consumption, making it ideal for highway environments.

(03) Circuit Design:

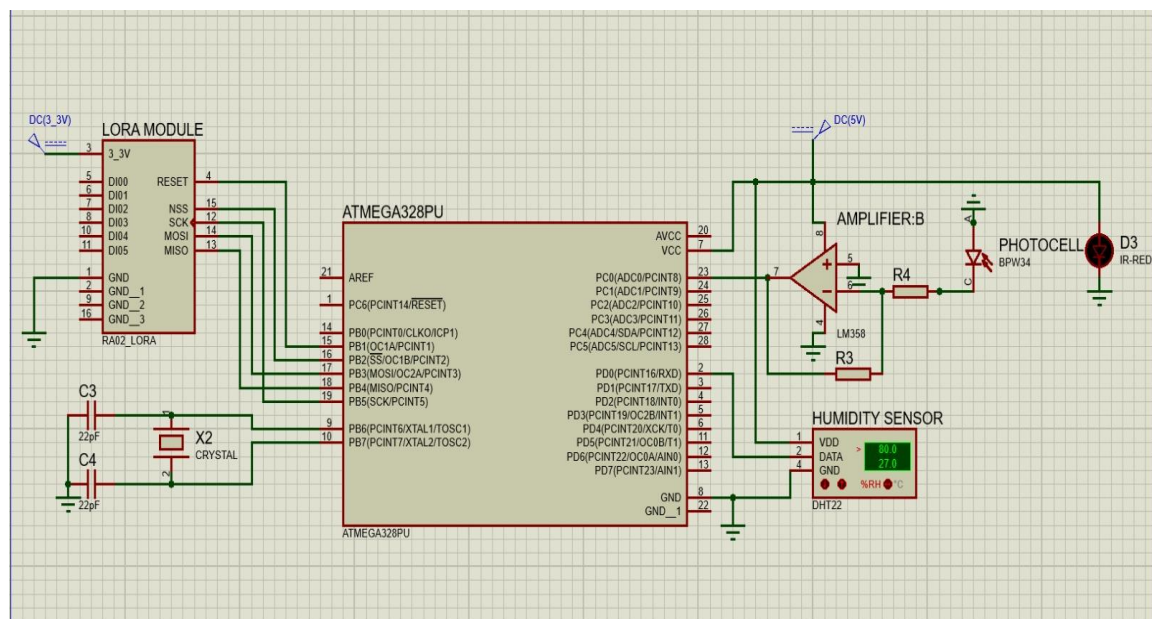


FIGURE 02 : TRANSMITTER SCHEMATIC DIAGRAM

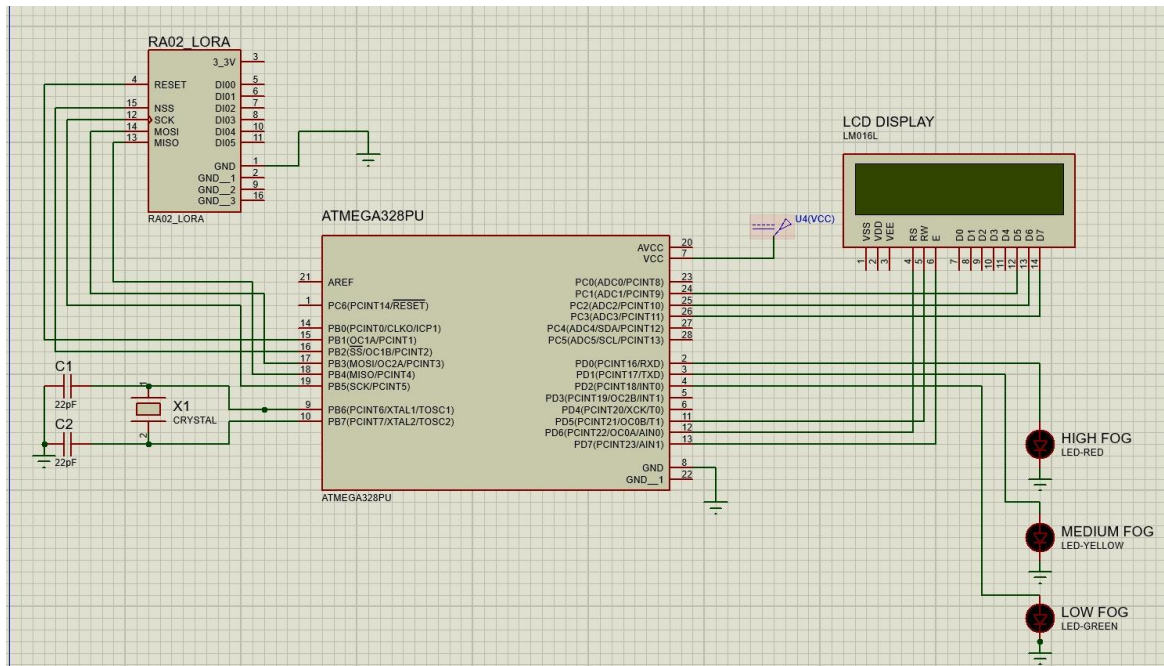


FIGURE 03 : RECEIVER SCHEMATIC DIAGRAM

HARDWARE COMPONENTS:

TABLE 01: COMPONENTS AND PURPOSES

Component	Purpose
ATmega328p Microcontroller	Main control unit
IR LED	Fog illumination
BPW34 Photocell	Detect fog density by measuring the attenuation and scattering of infrared (IR) light
LM358 Op-Amp	Amplify the Voltage given by BPW34
DHT22 Sensor	Measure humidity
RA-02 (SX1278) LoRa Module	Wireless communication
Resistance	Protect the components
LEDs	Fog level indication
16 MHz Crystal + Capacitors	Clock generation
Power Supply	5V & 3.3V supply

SOFTWARE TECHNOLOGIES:

TABLE 02:SOFTWARE TECHNOLOGIES AND USAGES

Technology	Usage
Embedded C	Firmware development
AVR-GCC	Compilation
Proteus	Circuit Designing
SPI Driver	LoRa communication

UNIQUENESS:

1. Innovation:

- Uses infrared scattering, not cameras.
- Works in low light, night, and fog.
- Low-cost and low-power design.
- Long-range communication using LoRa.
- Modular and scalable for multiple highway nodes.

2. Comparison with Existing Solutions

TABLE 03 : COMPARISON WITH EXISTING SOLUTION

Feature	Existing Camera Systems	FogGuard
Cost	High	Low
Night operation	Limited	Excellent
Processing	Heavy (image processing)	Lightweight
Power	High	Low
Communication	Wired / Cellular	LoRa (wireless)

BUDGET:

TABLE 04: BUDGET OF THE PROJECT

Component	Quantity	Cost
ATmega328p Microcontroller	2	1700.00
IR LED	1	20.00
BPW34 Photocell	1	300.00
LM358 Op-Amp	1	20.00
DHT22 Sensor	1	450.00
RA-02 (SX1278) LoRa Module	2	3400.00
Resistance and potentiometer	10	100.00
LEDs	3	15.00
16 MHz Crystal + Capacitors	1	30.00
Total cost		6055.00

TIMELINE:

TABLE 5: TIMELINE OF PROGRESS[illegible]